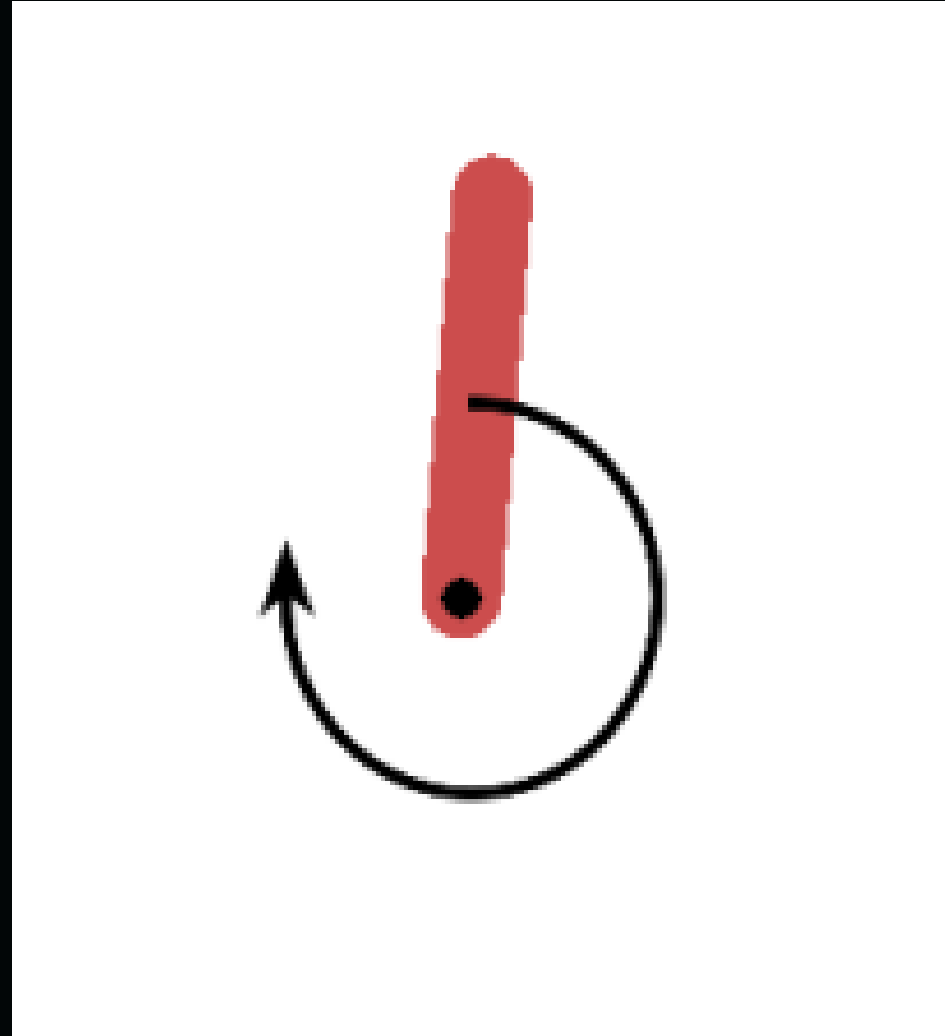


Part B: Pendulum Control with Deep Q-Network (DQN)

Goal: apply torque to swing up and stabilise the pendulum while maximising native Gym reward.



STATE
[$\cos(\theta)$, $\sin(\theta)$, θ]

ACTION
Continuous torque: -2 to +2

REWARD
Higher return = better balance

Required gravity settings

Default
10

Free-fall
0

Anti
-10

Super
15

Challenge: standard DQN needs discrete actions, but Pendulum provides continuous torque.

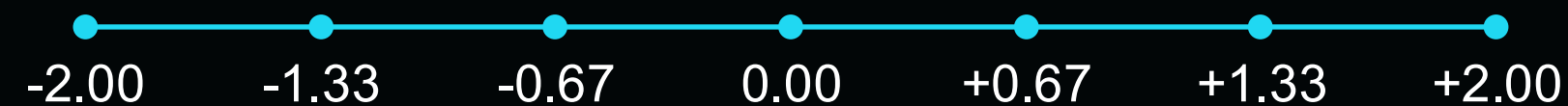
Why DQN Works Here

Discretise torque, then let the Q-network choose the best fixed torque action.



TORQUE DISCRETISATION

3, 5 and 7 actions were tested before architecture selection.
Final seven torques:

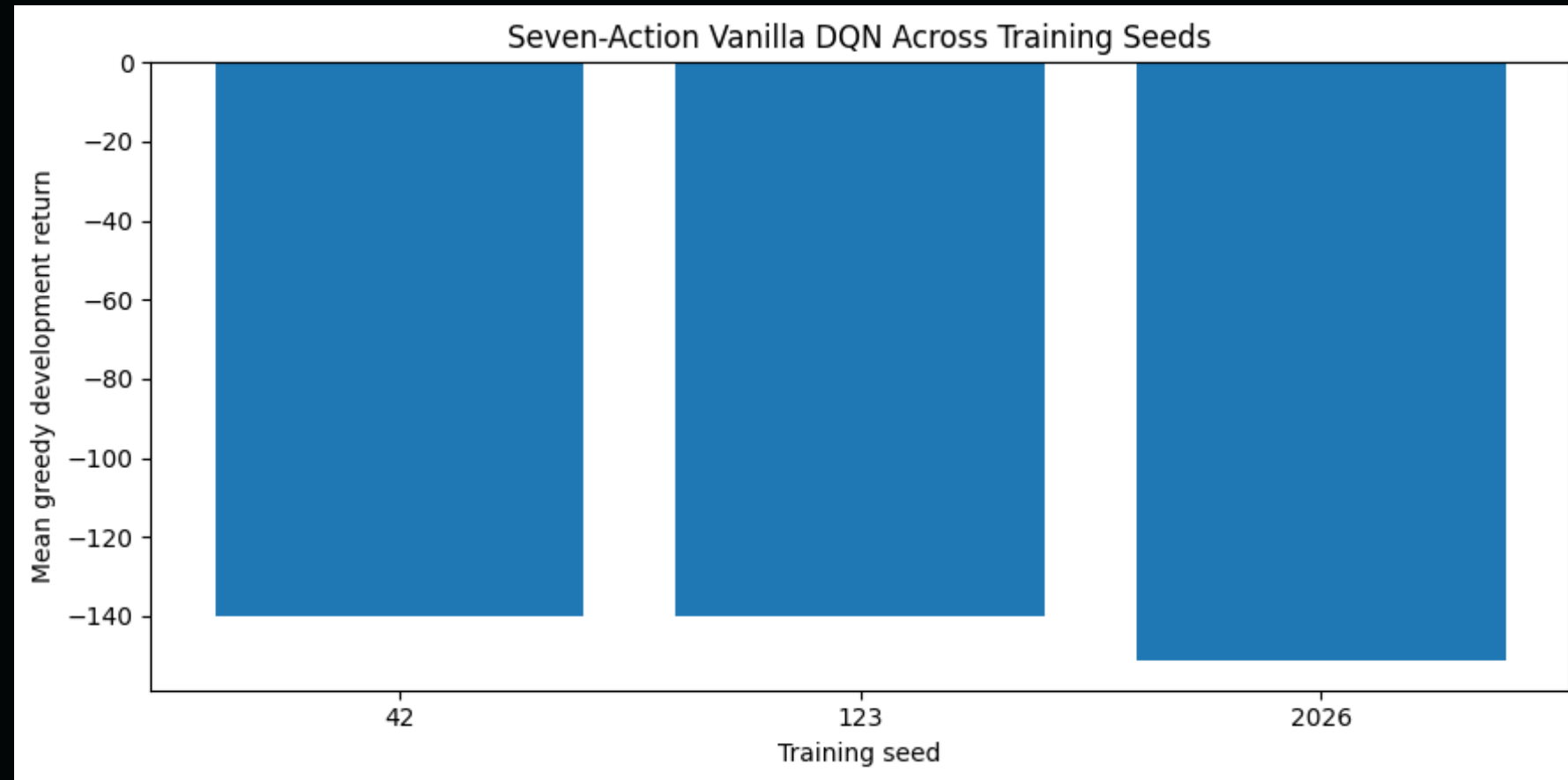


7 actions selected: mean development return = -139.82

Action set	Mean greedy development return	Difference vs 7 actions	Rank
3 actions	-140.077383	-0.258782	2
5 actions	-140.763848	-0.945247	3
7 actions	-139.818601	Best	1 ✓

Evaluation Protocol: How We Avoided Lucky Results

One RL run was not enough, so decisions used fixed states, matched seeds and a held-out final suite.



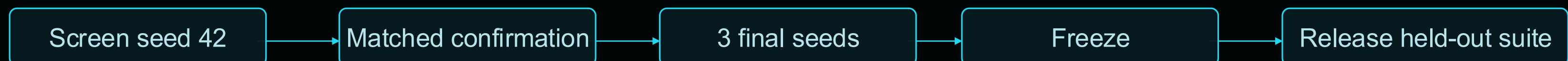
Vanilla seed means: 42 = -139.82 | 123 = -139.91 | 2026 = -151.15
Observed range = 11.34

Four evaluation rules

- 1. 24 fixed development states**
used for tuning and checkpoint selection
- 2. 24 separate held-out states**
opened only after the configuration was frozen
- 3. Matched states and seeds**
candidate and reference faced the same conditions
- 4. Greedy formal evaluation**
exploration disabled during final evaluation

Differences within 11.34 were treated cautiously and required matched-seed confirmation.

Decision pipeline



Systematic Optimisation: Why Dueling DQN Was Retained

We changed one design question at a time and retained only supported changes.

Experiment decision trail

Question tested

3 vs 5 vs 7 actions

Vanilla vs Double vs Dueling

Exploration schedule

Replay / target recipe

MSE vs Huber + clipping

gamma 0.95 vs 0.99

Raw vs scaled state

Decision

7 retained

Dueling retained

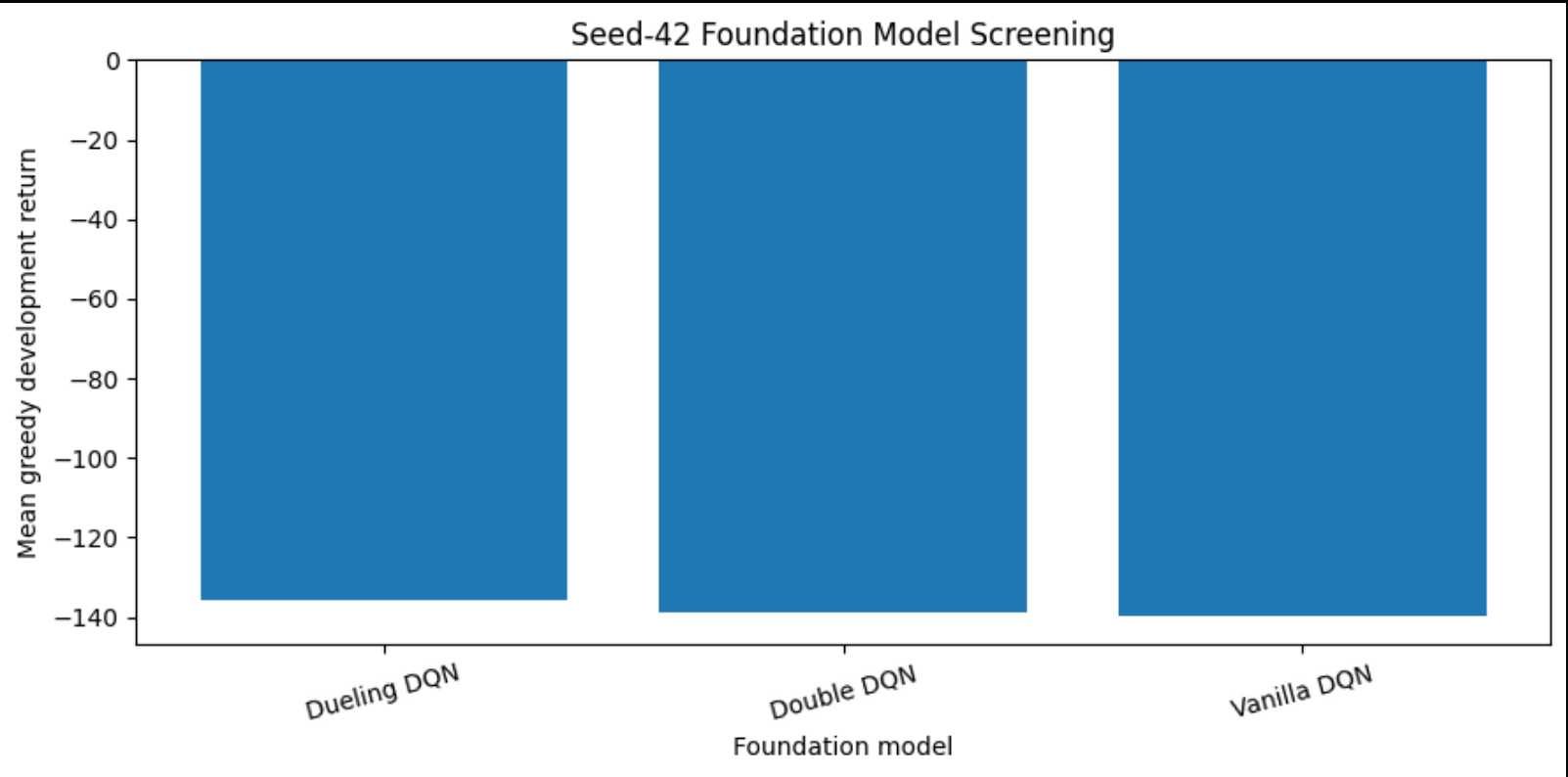
Moderate decay retained

Lecturer-aligned retained

MSE retained

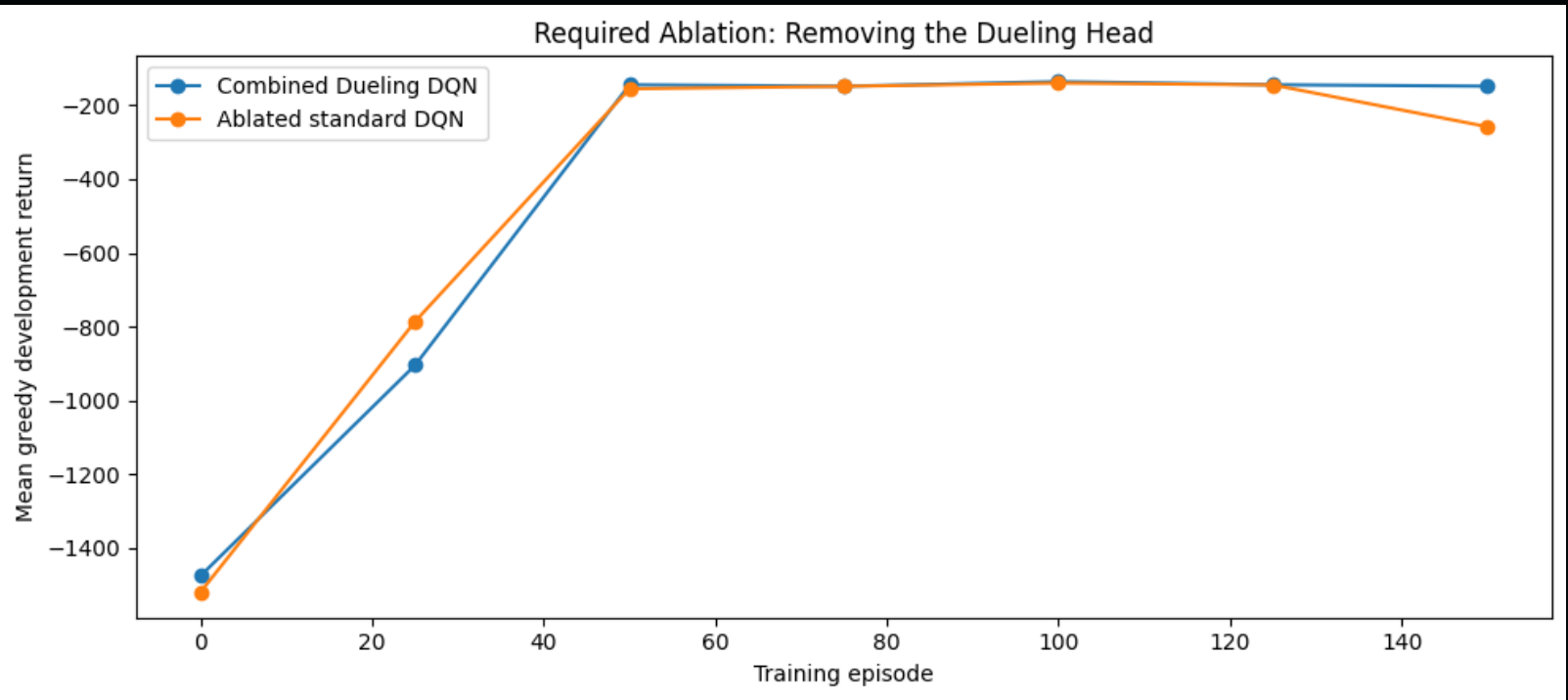
0.99 retained

Raw state retained



Matched Dueling vs Vanilla
Seed 42: +3.88 | Seed 123: +1.39
Average win rate: 81.25%

Ablation check
Dueling: -135.94 | Standard: -139.82
Extra parameters: only 65



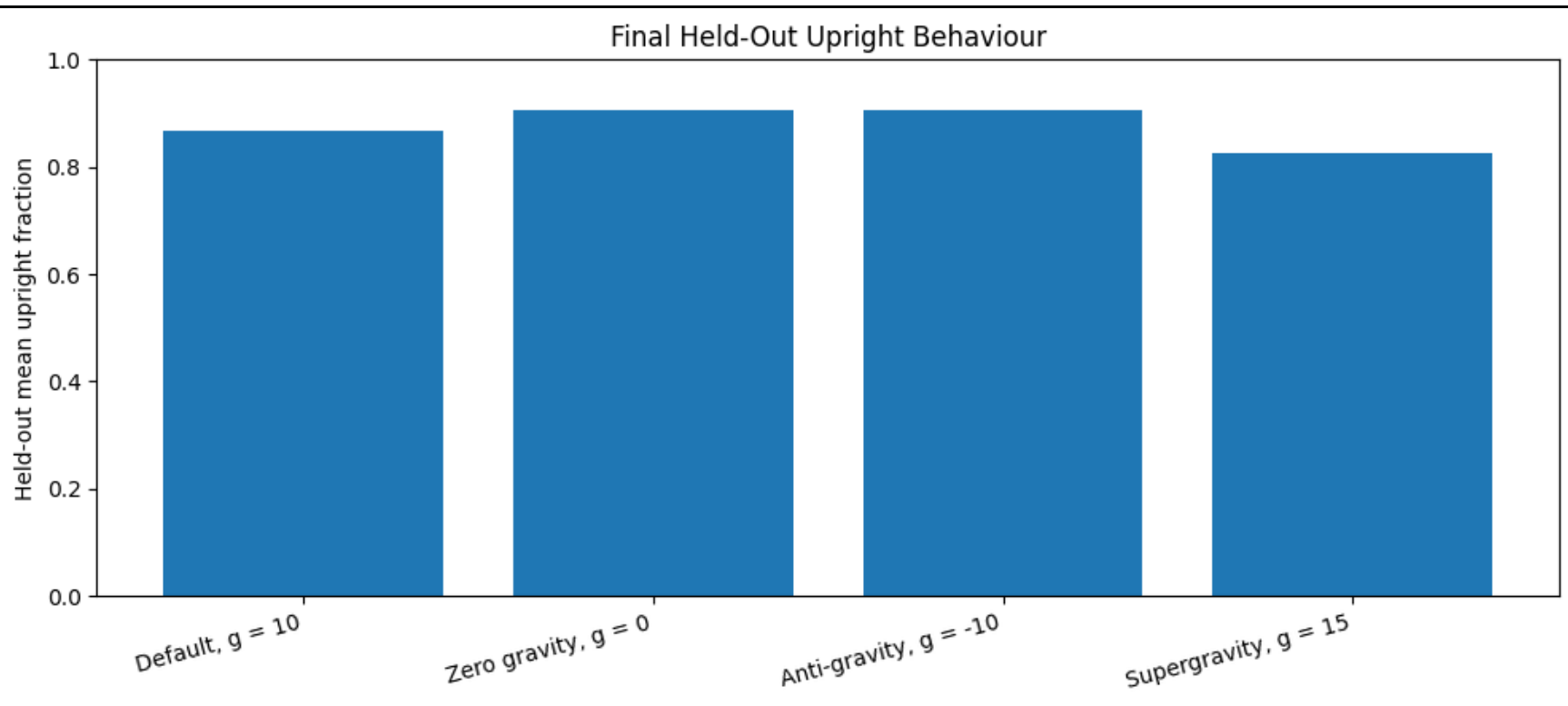
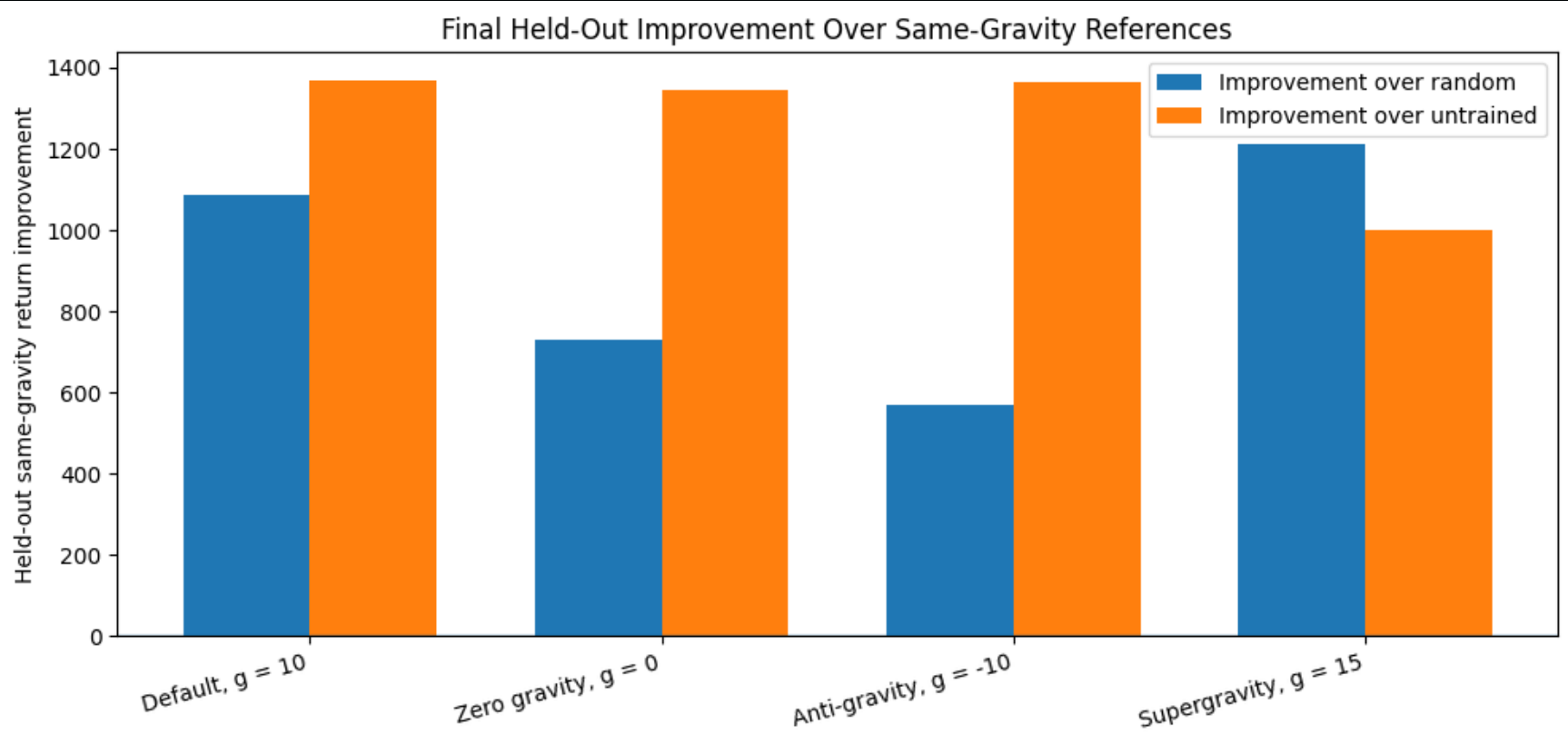
Most recipes tied: baseline performance was already strong by episodes 25 to 50.

Caveat: benefit was modest and within the 11.34 practical margin, so we do not claim statistical superiority.

Frozen Four-Gravity Test

Tuned under default gravity, then frozen and trained fresh under each required gravity.

Frozen setup: Dueling DQN | 64,64 hidden units | 7 torques | gamma 0.99 | 150 episodes | 30,000 steps | gravity seed 4242



Final held-out results

Gravity	Mean return	Upright
g = 10	-135.38	86.8%
g = 0	-53.77	90.7%
g = -10	-53.37	90.5%
g = 15	-203.31	82.7%

All four trained policies beat both same-gravity baselines.

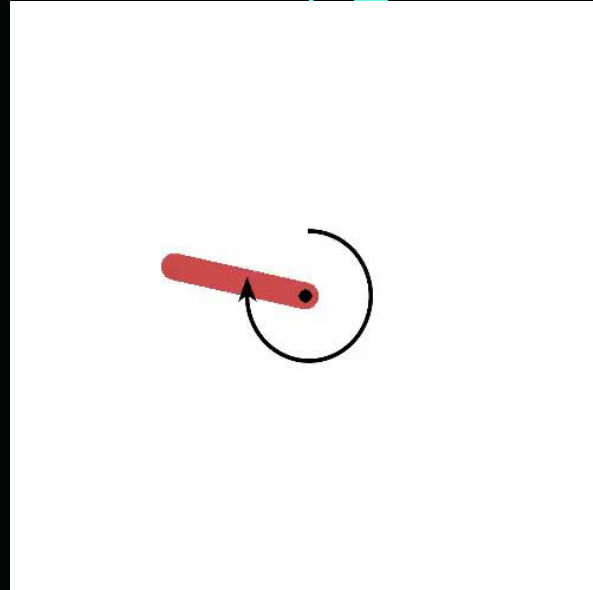
Do not rank raw returns directly across gravities because dynamics change.

Specific limitation
Supergravity showed the lowest upright fraction and clearest late-training degradation.

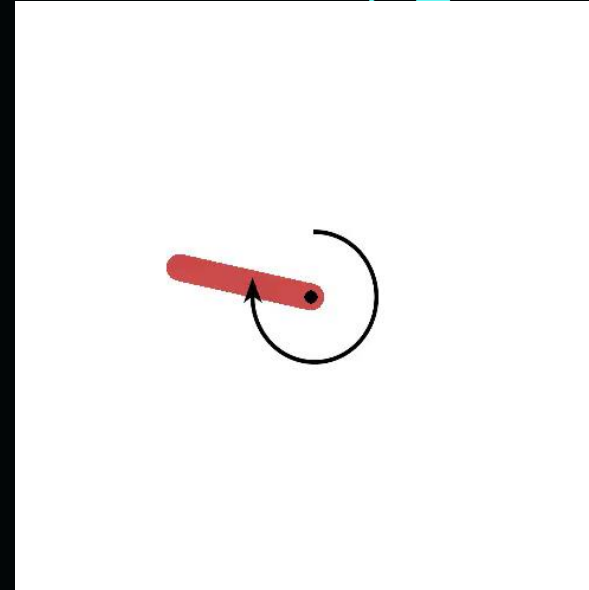
Demo, Reproducibility and Final Takeaway

Final videos use the same held-out starting state F08, not a cherry-picked trajectory.

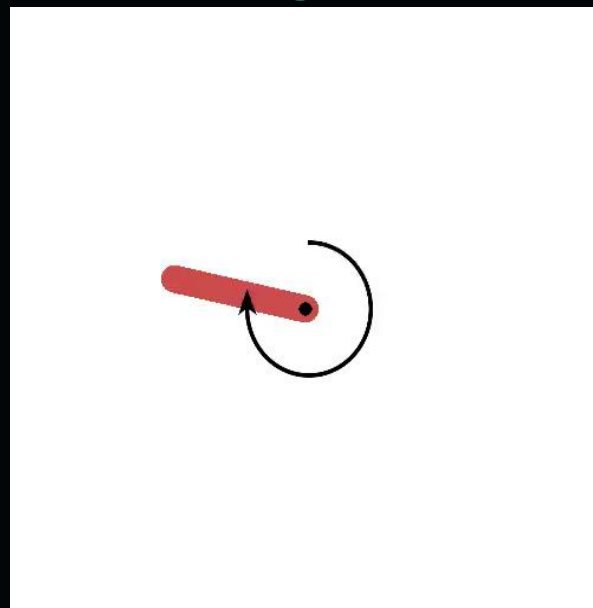
Default, $g = 10$



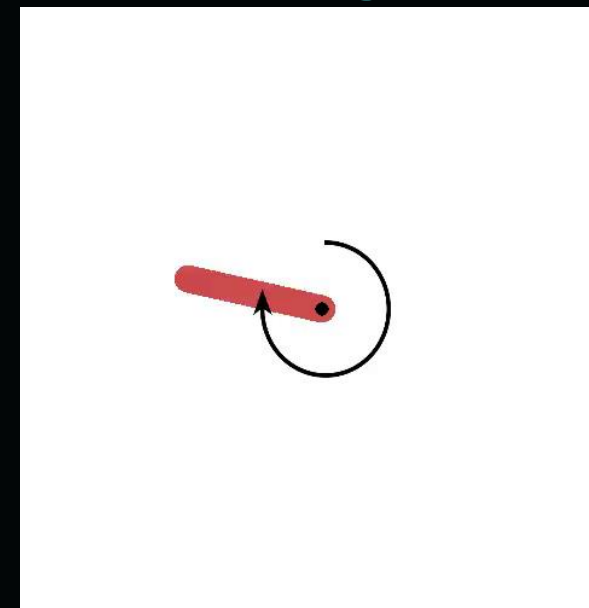
Free-fall, $g = 0$



Anti, $g = -10$



Super, $g = 15$



Final representative F08 videos

Fresh reload verification

4 / 4 final policies reproduced all held-out metrics

Mean-return difference = 0.0

Final conclusion

Dueling DQN with seven discrete torques was viable and reproducible under all four required gravity settings.

Limits

Discrete torque approximation, finite training budget, and one post-freeze seed for each non-default gravity.

Main takeaway

The result is backed by held-out metrics, reload checks and the four-video demo.